

Exercise 1 - SQL fundamentals

(Select & filtering statements)

Id	First name	Last name	Department	Salary	Hire-date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-01-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-12-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Question 1

Write a SQL query to retrieve all columns from the employees table

```
- SELECT *
  FROM employees
```

Question 2

Write a SQL query to find all unique departments in the employees table

```
- SELECT DISTINCT Department
  FROM employees
```


Departments

IT

HR

Finance

Marketing

Questions^x

Write a SQL query to retrieve all employees first and last names, ordered by Salary in descending order.

SELECT *

FROM table_name

WHERE Condition

ORDER BY Column_name; DESC

SELECT

first_name

last_name

Salary⁸

FROM Employees

ORDER BY Salary^{DESC}

First_name	Last_name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000

Session	Room	Capacity
Lecture	Moore	5000
Lecture	Hall	5000
Lecture	Smith	1000

Question 4

Write a SQL query to retrieve the top 6 highest-paid employees.

```
SELECT *
FROM Employees
Order By Salary Desc
Limit 6;
```

Salary
62000
60000
59000
55000
53000

Question 5

Write a SQL query to find employees who work in the IT department.

```
SELECT *
FROM Employees
Where Department = IT
```


Question 1

SELECT *

FROM employees

Where department not in IT

ID	First-name	Last-name	Department	Salary	Hire-date	City
01	Jane	Smith	HR	71000	2017-02-20	Chicago
3	Mike	Johnson	Finance	60000	2011-09-20	Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	59000	2017-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2012-06-01	Chicago

Question 1

SELECT *

FROM employees

Where Department in (HR, IT or Finance)

First-name	Last-name	Department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

Question 10

SELECT ✓

FROM Employees

Where department = IT

AND Salary > 50 000

AND City = New York

ID	First name	Last name	Department	Salary	City
1	John	Doe	IT	55 000	New York
4	Sarah	Brown	IT	53 000	New York

Question 11 ✓SELECT first_name,
Last_name

FROM Employees

Where department = (Finance or Marketing)

AND Salary > 52 000

Order By Salary desc

First name	Last name	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

Question 12 ✓

SELECT DISTINCT City
 FROM Employees
 Where department not in (IT, HR),

City
Los Angeles
San Francisco
Houston
Chicago

Question 13 ✓

SELECT *
 FROM employees
 Where department <> Finance
 AND Salary > 50000
 Order By Hire_date ASC,

first_name	last_name	Department	Salary	Hire_date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2016-05-22
John	Doc	IT	55000	2015-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

Question 2nd

SELECT * first_name, last_name, City, Department
 FROM EMPLOYEES
 WHERE City IN (Chicago, Los Angeles)
 AND Department IN (IT, Marketing)
 Limit 3;

First_name	Last_name	Department	City
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago